

SULIT



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SCHOOL OF COMPUTING
Faculty of Engineering

UNIVERSITI TEKNOLOGI MALAYSIA
FINAL EXAMINATION SEMESTER I, 2018 / 2019

SUBJECT CODE : SCSI 2143
SUBJECT NAME : PROBABILITY & STATISTICAL DATA ANALYSIS
SECTION : ALL
TIME : 9.00 AM – 12.00 NOON
DATE/DAY : 03 / 01 / 2019 (THURSDAY)
VENUES : BK 2 (N 28)

INSTRUCTIONS :

Answer **ALL** question in the answer booklet provided.

Provide your final answer accurate to 3 decimal places.

(Please Write Your Lecturer Name And Section In Your Answer Booklet)

Name	
I/C No.	
Year / Course	
Section	
Lecturer Name	

This question paper consists of **ELEVEN (11)** printed pages including this page.

Question 1**[25 Marks]**

- a) Seven housewives were randomly sampled, and it was determined that they walked an average of 39.2 km per week in their housework, with a sample standard deviation of 3.23 km per week. Construct a 95 percent confident interval for the population mean of walked per week of housewives in their housework. (7 marks)
- b) Hinton Press hypothesizes that the life of its largest web press is 14,500 hours, with a known standard deviation of 2,800 hours. From a sample of 25 presses, the company finds a sample mean of 13,000 hours. At a 0.01 significance level, should the company conclude that the average life of the presses is less than the hypothesized 14,500 hours? (10 marks)
- c) A standard deviation of a distribution is hypothesized to be 50. If an observed sample of 30 yield a sample standard deviation of 57, should we reject the null hypothesis that the true standard deviation is 50? Use the 0.05 level of significance. (8 marks)

Question 2**[20 Marks]**

- a) A ketchup manufacturer is in the process of deciding whether to produce a new extra spicy brand. The company's marketing-research department used a national telephone survey of 6,000 households and found that the extra-spicy ketchup would be purchased by 335 of them. A much extensive study made 2 years ago showed that 300 of the 6,000 households would purchase the brand then. At a 0.02 significance level, should the company conclude that there is an increased interest in the extra-spicy flavour? (10 marks)
- b) A private university has just introduced a new e-learning programme. The university randomly selected 10 students as its research subjects to study whether the programme is effective in improving their performance. The performance score before and after the e-learning program are shown in Table 1:

Table 1 : Data of student performance

Student	Performance score before the e-learning programme	Performance score after the e-learning programme
1	66	68
2	34	40
3	67	66
4	88	89
5	56	60
6	75	76
7	88	90
8	66	66
9	55	58
10	45	34

- i. Determine the effectiveness of the e-learning programme by using a suitable t-test at a 0.05 significance level. (8 marks)
- ii. Report the results of the analysis. (2 marks)

Question 3

[20 Marks]

- a) A new donut shop plans to sell plain, strawberry, blueberry and cinnamon donuts. They wonder if there is a preference for one of these types of donuts or if each type is preferred by the same proportion of customers. The data of the customer preference is shown in Table 2 :

Table 2 : Data of customer preference

	Plain	Strawberry	Blueberry	cinnamon
Observed count	13	12	16	19

- i. State the null and alternative hypothesis (2 marks)
 - ii. Find the critical value (3 marks)
 - iii. Calculate the test statistics (3 marks)
 - iv. What does the result suggest about the claim that customer preference on each type of donut is similar? (2 marks)
- b) A movie theatre recorded the type of snack item purchased and the type of movie the customer was attending for each person in a random sample of theatre customer who made a snack bar purchase. The resulting data as shown in the Table 3. At the 0.05 significant level, test the claim that, type of movies independent with type of snack sold (10 marks)

Table 3 : Type of movie with type of snack sold

	Soda	Popcorn	candy
Children's movie	70	83	47
Action movie	61	49	20

Question 4

[20 Marks]

- a) Determine the correlation coefficient for the data in Table 4. Then explain the relationship between thinking skills and IQ based on the correlation coefficient.

(10 marks)

Table 4 : Data on relationship between thinking skills and IQ

Subject	Thinking skills score	IQ score
1	92	98
2	60	65
3	72	78
4	54	53
5	88	90
6	95	100
7	79	82
8	72	76
9	58	58
10	89	95

- b) Zippy Cola is studying the effect of its latest advertising campaign. People chosen at random, were called and asked how many cans of Zippy Cola they had bought in the past week and how many Zippy Cola advertisements they had either read or seen in the past week. The data from their study are presented in Table 5:

Table 5 : Data on number of advertisements and number of Zippy Cola purchased

X (number of ads)	3	7	4	2	0	4	1	2
Y(can purchased)	11	18	9	4	7	6	3	8

Develop the regression equation that best fits the data.

(10 marks)

Question 5**[15 Marks]**

Students were given different drug treatment before revising for their exam. Some were given a memory drug, some a placebo drug and some no treatment. The exam score (%) are shown below for the three different group. Test the hypotheses that the treatments will have different effect on exam score by using $\alpha = 0.05$. (15 marks)

Table 6 : Memory drug for different group

<i>n</i>	Memory drug	placebo	No treatment
1	70	37	3
2	77	43	10
3	83	50	17
4	90	57	23
5	97	63	30
mean	83.4	50	16.6
variance	112.3	109.0	112.3

List of Formula:

(a) Interval Estimation

$$\bar{x} \pm (Z_{\frac{\alpha}{2}}) \left(\frac{\sigma}{\sqrt{n}} \right)$$

$$\bar{x} \pm (t_{\alpha, dof}) \left(\frac{s}{\sqrt{n}} \right)$$

(b) Hypothesis Testing for one sample

$$\text{Test Statistic, } z_0 = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

$$\text{Test Statistic, } t_0 = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

$$\text{Test Statistic, } \chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

(c) Hypothesis Testing for two samples

$$\text{Test Statistic, } z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

$$\text{With } \hat{p}_1 = \frac{x_1}{n_1}, \hat{p}_2 = \frac{x_2}{n_2}, \bar{p} = \frac{x_1 + x_2}{n_1 + n_2}, \bar{q} = 1 - \bar{p}$$

$$\text{Test Statistic, } z_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

$$\text{Test Statistic, } t_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}; s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{x}_1 - \bar{x}_2 - \Delta_0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; v = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}$$

$$\text{Test Statistic, } T_0 = \frac{\bar{D} - u_D}{\frac{S_D}{\sqrt{n}}}$$

$$\text{Test Statistic, } F_0 = \frac{s_1^2}{s_2^2}$$

$$S_D = \sqrt{\frac{\sum D^2 - \frac{(\sum D)^2}{n}}{n-1}}$$

(d) Chi-Square Test

$$\text{Test Statistic, } \chi^2 = \sum_i^n \frac{(o_i - e_i)^2}{e_i}$$

$$\text{Degree of Freedom for two-way contingency table} = (r - 1)(c - 1)$$

(e) Correlation

$$\text{Pearson Correlation Coefficient, } r = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sqrt{\left(\sum x^2 - \frac{(\sum x)^2}{n}\right) \left(\sum y^2 - \frac{(\sum y)^2}{n}\right)}}$$

$$\text{Spearman Correlation Coefficient, } r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \text{ where } d_i = y_i - x_i$$

(f) Regression

$$b_1 = \frac{\sum xy - \frac{(\sum x \sum y)}{n}}{\sum x^2 - \frac{(\sum x)^2}{n}}$$

$$b_0 = \bar{y} - b_1 \bar{x}$$

(g) Analysis of variance (ANOVA)

$$\text{Test Statistic, } F = \frac{ns^{\frac{2}{x}}}{s_p^2}$$

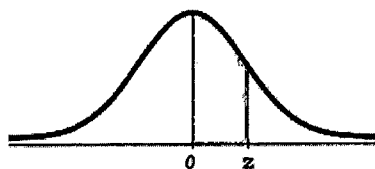


TABLE A-2 Standard Normal (z) Distribution										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2828	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495 *	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591 ▲	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949 *	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962 ▲	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990
3.10 and higher	.4999									

NOTE: For values of z above 3.09, use 0.4999 for the area.

*Use these common values that result from interpolation:

z score Area

1.645 0.4500 ←

2.575 0.4950 ←

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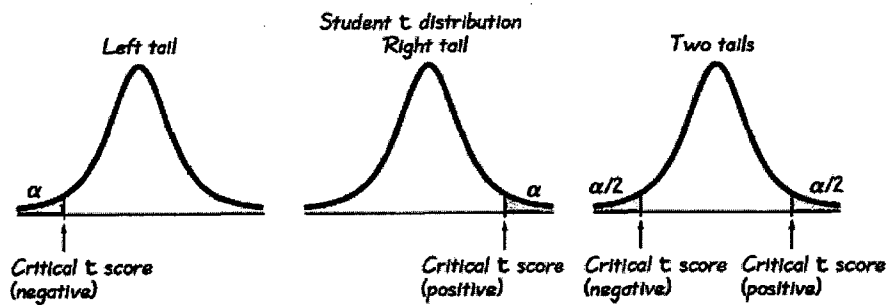


TABLE A-3 t Distribution

Degrees of Freedom	α					
	.005 (one tail) .01 (two tails)	.01 (one tail) .02 (two tails)	.025 (one tail) .05 (two tails)	.05 (one tail) .10 (two tails)	.10 (one tail) .20 (two tails)	.25 (one tail) .50 (two tails)
1	63.657	31.821	12.706	6.314	3.078	1.000
2	9.925	6.965	4.303	2.920	1.886	.816
3	5.841	4.541	3.182	2.353	1.638	.765
4	4.604	3.747	2.776	2.132	1.533	.741
5	4.032	3.365	2.571	2.015	1.476	.727
6	3.707	3.143	2.447	1.943	1.440	.718
7	3.500	2.998	2.365	1.895	1.415	.711
8	3.355	2.896	2.306	1.860	1.397	.706
9	3.250	2.821	2.262	1.833	1.383	.703
10	3.169	2.764	2.228	1.812	1.372	.700
11	3.106	2.718	2.201	1.796	1.363	.697
12	3.054	2.681	2.179	1.782	1.356	.696
13	3.012	2.650	2.160	1.771	1.350	.694
14	2.977	2.625	2.145	1.761	1.345	.692
15	2.947	2.602	2.132	1.753	1.341	.691
16	2.921	2.584	2.120	1.746	1.337	.690
17	2.898	2.567	2.110	1.740	1.333	.689
18	2.878	2.552	2.101	1.734	1.330	.688
19	2.861	2.540	2.093	1.729	1.328	.688
20	2.845	2.528	2.086	1.725	1.325	.687
21	2.831	2.518	2.080	1.721	1.323	.686
22	2.819	2.508	2.074	1.717	1.321	.686
23	2.807	2.500	2.069	1.714	1.320	.685
24	2.797	2.492	2.064	1.711	1.318	.685
25	2.787	2.485	2.060	1.708	1.316	.684
26	2.779	2.479	2.056	1.706	1.315	.684
27	2.771	2.473	2.052	1.703	1.314	.684
28	2.763	2.467	2.048	1.701	1.313	.683
29	2.756	2.462	2.045	1.699	1.311	.683
Large (z)	2.575	2.326	1.960	1.645	1.282	.675

TABLE A-4 Chi-Square (χ^2) Distribution

Degrees of Freedom	Area to the Right of the Critical Value									
	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
1	—	—	0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.054	0.103	0.211	3.841	5.024	5.974	7.378	7.879
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.142	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.071	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.597	12.592	14.449	16.812	18.549
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.312	1.646	2.180	2.716	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.936	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.076	3.572	4.403	5.229	6.311	18.575	21.026	23.337	26.217	28.306
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.628	6.578	7.790	21.064	23.685	26.119	29.141	31.319
15	4.601	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578	32.801
16	5.142	5.812	6.913	7.962	9.312	23.542	26.296	28.845	32.000	34.267
17	5.697	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409	35.718
18	6.268	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805	37.156
19	6.844	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191	38.582
20	7.434	8.261	9.591	10.857	12.443	28.412	31.410	34.170	37.566	39.997
21	8.034	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932	41.401
22	8.643	9.549	10.982	12.338	14.042	30.813	33.924	36.781	40.289	42.796
23	9.260	10.196	11.689	13.091	14.848	32.007	35.172	38.076	41.638	44.181
24	9.886	10.856	12.401	13.848	15.659	33.196	36.415	39.364	42.980	45.559
25	10.520	11.524	13.120	14.611	16.473	34.382	37.652	40.646	44.314	46.928
26	11.160	12.198	13.845	15.379	17.290	35.567	38.885	41.924	45.645	48.289
27	11.808	12.879	14.573	16.151	18.114	36.741	40.113	43.194	46.963	49.645
28	12.461	13.568	15.308	16.928	18.944	37.916	41.337	44.461	48.278	50.994
29	13.121	14.257	16.047	17.708	19.768	39.087	42.557	45.722	49.588	52.336
30	13.787	14.953	16.791	18.492	20.599	40.256	43.773	46.979	50.892	53.672
40	20.707	22.164	24.433	26.509	29.051	51.805	55.758	59.342	63.691	66.766
50	27.991	29.819	32.357	34.264	37.689	63.691	67.505	71.420	76.154	79.490
60	35.534	37.485	40.482	43.188	46.459	74.397	79.082	83.298	88.379	91.952
70	43.275	45.442	48.758	51.759	55.329	85.527	90.531	95.023	100.425	104.215
80	51.172	53.540	57.153	60.391	64.278	96.578	101.879	106.629	112.329	116.321
90	59.342	61.751	65.657	69.296	73.291	107.565	112.916	117.975	123.890	128.009
100	67.328	70.065	74.222	77.929	82.358	118.498	124.342	129.561	135.807	140.169

F Values for $\alpha = 0.05$

d_2	d_1								
	1	2	3	4	5	6	7	8	9
1	161.4	199.5	215.7	224.6	230.2	234.0	236.8	238.9	240.5
2	18.51	19.00	19.16	19.25	19.3	19.33	19.35	19.37	19.38
3	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85	8.81
4	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04	6.00
5	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82	4.77
6	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15	4.10
7	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73	3.68
8	5.32	4.46	4.07	3.84	3.69	3.58	3.50	3.44	3.39
9	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.23	3.18
10	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.07	3.02
11	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.95	2.90
12	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.85	2.80
13	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.77	2.71
14	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.70	2.65
15	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.64	2.59
16	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.59	2.54
17	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.55	2.49
18	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51	2.46
19	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48	2.42
20	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45	2.39
21	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42	2.37
22	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40	2.34
23	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37	2.32
24	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36	2.30
25	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34	2.28
26	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32	2.27
27	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31	2.25
28	4.20	3.34	2.95	2.71	2.56	2.45	2.36	2.29	2.24
29	4.18	3.33	2.93	2.70	2.55	2.43	2.35	2.28	2.22
30	4.17	3.32	2.92	2.69	2.53	2.42	2.33	2.27	2.21
40	4.08	3.23	2.84	2.61	2.45	2.34	2.25	2.18	2.12
60	4.00	3.15	2.76	2.53	2.37	2.25	2.17	2.10	2.04
120	3.92	3.07	2.68	2.45	2.29	2.17	2.09	2.02	1.96
inf	3.84	3.00	2.60	2.37	2.21	2.10	2.01	1.94	1.88